Ex8 Report | Building a Web API

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Source code

See the attached source code with this submission.

This web API is built using NodeJS and MongoDB.

The project is structured as follows:

```
/project-folder
  |-- /models
  |     |-- sensorData.js
  |     |-- user.js
  |-- /routes
  |     |-- sensorRoutes.js
  |     |-- userRoutes.js
  |-- app.js
  |-- swagger.js (optional)
```

Models

I have 2 models for the sensor data and for user. We specify them as how they are described in the problem statement.

sensorData.js

```
const mongoose = require('mongoose');

const sensorDataSchema = new mongoose.Schema({
    Id: { type: Number, required: true },
    SensorName: { type: String, required: true },
    SensorValue: { type: Number, required: true },
})

module.exports = mongoose.model('SensorData', sensorDataSchema);
```

user.js

```
const mongoose = require('mongoose');

const userSchema = new mongoose.Schema({
    Id: { type: Number, required: true }, // Custom Id field
    UserName: { type: String, required: true },
    Password: { type: String, required: true },
    Email: { type: String, required: true },
});

module.exports = mongoose.model('User', userSchema);
```

Routes

We need to define the route (aka. Endpoints) for interacting with this API. Here I also include the Swagger documentation, but that's optional.

sensorRoutes.js

```
const express = require('express');
const router = express.Router();
const SensorData = require('../models/sensorData');

/**
 * @swagger
 * tags:
 *   name: SensorDatas
 *   description: API for managing SensorData
 */

/**
 * @swagger
 * /api/SensorDatas:
 *   get:
 *     summary: Get all SensorDatas
 *     tags: [SensorDatas]
 *     responses:
 *       200:
 *         description: A list of SensorDatas
 *         content:
 *           application/json:
 *             schema:
 *               type: array
 *               items:
 *                 $ref: '#/components/schemas/SensorData'
 */
```

```
router.get('/', async (req, res) => {
    try {
        const sensors = await SensorData.find();
        res.json(sensors);
    } catch (err) {
        res.status(500).json({ message: err.message });
    }
});

/**
 * @swagger
 * /api/SensorDatas:
 *   post:
 *     summary: Create a new SensorData
 *     tags: [SensorDatas]
 *     requestBody:
 *       required: true
 *       content:
 *         application/json:
 *           schema:
 *             $ref: '#/components/schemas/SensorData'
 *     responses:
 *       201:
 *         description: SensorData created successfully
 *         content:
 *           application/json:
 *             schema:
 *               $ref: '#/components/schemas/SensorData'
 *       400:
 *         description: Bad Request
 */

router.post('/', async (req, res) => {
    const sensor = new SensorData({
        Id: req.body.Id, // Ensure unique ID
        SensorName: req.body.SensorName,
        SensorValue: req.body.SensorValue,
    });

    try {
        const newSensor = await sensor.save();
        res.status(201).json(newSensor);
    } catch (err) {
        res.status(400).json({ message: err.message });
    }
});
```

```
/**
 * @swagger
 * /api/SensorDatas/{id}:
 *   get:
 *     summary: Get a SensorData by ID
 *     tags: [SensorDatas]
 *     parameters:
 *       - in: path
 *         name: id
 *         required: true
 *         schema:
 *           type: integer
 *         description: The SensorData ID
 *     responses:
 *       200:
 *         description: SensorData retrieved successfully
 *         content:
 *           application/json:
 *             schema:
 *               $ref: '#/components/schemas/SensorData'
 *       404:
 *         description: SensorData not found
 */
router.get('/:id', async (req, res) => {
    try {
        const sensor = await SensorData.findOne({ Id: req.params.id });
        if (!sensor) return res.status(404).json({ message: 'Sensor not
found' });
        res.json(sensor);
    } catch (err) {
        res.status(500).json({ message: err.message });
    }
});

/**
 * @swagger
 * /api/SensorDatas/{id}:
 *   put:
 *     summary: Update a SensorData by ID
 *     tags: [SensorDatas]
 *     parameters:
 *       - in: path
 *         name: id
 *         required: true
 *         schema:
 *           type: integer
 *         description: The SensorData ID
 *     requestBody:
```

```
 *       required: true
 *       content:
 *         application/json:
 *           schema:
 *             $ref: '#/components/schemas/SensorData'
 *     responses:
 *       200:
 *         description: SensorData updated successfully
 *         content:
 *           application/json:
 *             schema:
 *               $ref: '#/components/schemas/SensorData'
 *       404:
 *         description: SensorData not found
 *       400:
 *         description: Bad Request
 */
router.put('/:id', async (req, res) => {
    try {
        const updatedSensor = await SensorData.findOneAndUpdate(
            { Id: req.params.id },
            req.body,
            { new: true }
        );
        if (!updatedSensor) return res.status(404).json({ message: 'Sensor
not found' });
        res.json(updatedSensor);
    } catch (err) {
        res.status(400).json({ message: err.message });
    }
});

/**
 * @swagger
 * /api/SensorDatas/{id}:
 *   delete:
 *     summary: Delete a SensorData by ID
 *     tags: [SensorDatas]
 *     parameters:
 *       - in: path
 *         name: id
 *         required: true
 *         schema:
 *           type: integer
 *         description: The SensorData ID
 *     responses:
 *       200:
 *         description: SensorData deleted successfully
```

```
 *       404:
 *         description: SensorData not found
 */
router.delete('/:id', async (req, res) => {
    try {
        const sensor = await SensorData.findOneAndDelete({ Id:
req.params.id });
        if (!sensor) return res.status(404).json({ message: 'Sensor not
found' });
        res.json({ message: 'Sensor deleted' });
    } catch (err) {
        res.status(500).json({ message: err.message });
    }
});

module.exports = router;
```

userRoutes.js

```
const express = require('express');
const router = express.Router();
const User = require('../models/user');

/**
 * @swagger
 * tags:
 *   name: Users
 *   description: API for managing Users
 */

/**
 * @swagger
 * /api/Users:
 *   get:
 *     summary: Get all Users
 *     tags: [Users]
 *     responses:
 *       200:
 *         description: A list of Users
 *         content:
 *           application/json:
 *             schema:
 *               type: array
 *               items:
 *                 $ref: '#/components/schemas/User'
 */
```

```
router.get('/', async (req, res) => {
    try {
        const users = await User.find();
        res.json(users);
    } catch (err) {
        res.status(500).json({ message: err.message });
    }
});

/**
 * @swagger
 * /api/Users:
 *   post:
 *     summary: Create a new User
 *     tags: [Users]
 *     requestBody:
 *       required: true
 *       content:
 *         application/json:
 *           schema:
 *             $ref: '#/components/schemas/User'
 *     responses:
 *       201:
 *         description: User created successfully
 *         content:
 *           application/json:
 *             schema:
 *               $ref: '#/components/schemas/User'
 *       400:
 *         description: Bad Request
 */
router.post('/', async (req, res) => {
    const user = new User({
        Id: req.body.Id, // Ensure unique ID
        UserName: req.body.UserName,
        Password: req.body.Password,
        Email: req.body.Email,
    });

    try {
        const newUser = await user.save();
        res.status(201).json(newUser);
    } catch (err) {
        res.status(400).json({ message: err.message });
    }
});

/**
```

```
 * @swagger
 * /api/Users/{id}:
 *   get:
 *     summary: Get a User by ID
 *     tags: [Users]
 *     parameters:
 *       - in: path
 *         name: id
 *         required: true
 *         schema:
 *           type: integer
 *         description: The User ID
 *     responses:
 *       200:
 *         description: User retrieved successfully
 *         content:
 *           application/json:
 *             schema:
 *               $ref: '#/components/schemas/User'
 *       404:
 *         description: User not found
 */
router.get('/:id', async (req, res) => {
    try {
        const user = await User.findOne({ Id: req.params.id });
        if (!user) return res.status(404).json({ message: 'User not
found' });
        res.json(user);
    } catch (err) {
        res.status(500).json({ message: err.message });
    }
});

/**
 * @swagger
 * /api/Users/{id}:
 *   put:
 *     summary: Update a User by ID
 *     tags: [Users]
 *     parameters:
 *       - in: path
 *         name: id
 *         required: true
 *         schema:
 *           type: integer
 *         description: The User ID
 *     requestBody:
 *       required: true
```

```
 *       content:
 *         application/json:
 *           schema:
 *             $ref: '#/components/schemas/User'
 *     responses:
 *       200:
 *         description: User updated successfully
 *         content:
 *           application/json:
 *             schema:
 *               $ref: '#/components/schemas/User'
 *       404:
 *         description: User not found
 *       400:
 *         description: Bad Request
 */
router.put('/:id', async (req, res) => {
    try {
        const updatedUser = await User.findOneAndUpdate(
            { Id: req.params.id },
            req.body,
            { new: true }
        );
        if (!updatedUser) return res.status(404).json({ message: 'User not
found' });
        res.json(updatedUser);
    } catch (err) {
        res.status(400).json({ message: err.message });
    }
});

/**
 * @swagger
 * /api/Users/{id}:
 *   delete:
 *     summary: Delete a User by ID
 *     tags: [Users]
 *     parameters:
 *       - in: path
 *         name: id
 *         required: true
 *         schema:
 *           type: integer
 *         description: The User ID
 *     responses:
 *       200:
 *         description: User deleted successfully
 *       404:
```

```
 *         description: User not found
 */
router.delete('/:id', async (req, res) => {
    try {
        const user = await User.findOneAndDelete({ Id: req.params.id });
        if (!user) return res.status(404).json({ message: 'User not
found' });
        res.json({ message: 'User deleted' });
    } catch (err) {
        res.status(500).json({ message: err.message });
    }
});

module.exports = router;
```

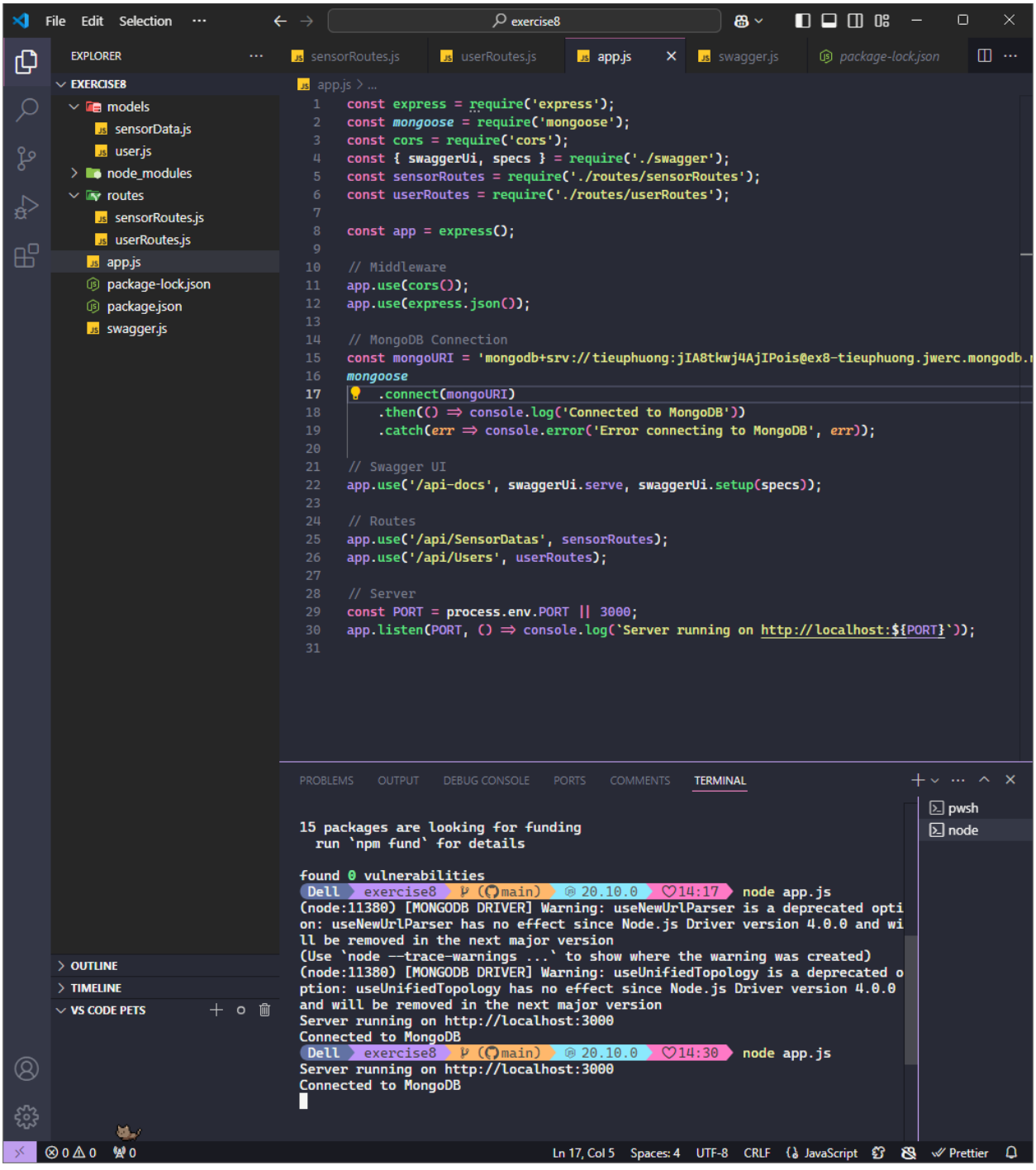
Main Application

App.js

```
const express = require('express');
const mongoose = require('mongoose');
const cors = require('cors');
const { swaggerUi, specs } = require('./swagger');
const sensorRoutes = require('./routes/sensorRoutes');
const userRoutes = require('./routes/userRoutes');

const app = express();

// Middleware
app.use(cors());
app.use(express.json());

// MongoDB Connection
const mongoURI = 'mongodb+srv://tieuphuong:jIA8tkwj4AjIPois@ex8-
tieuphuong.jwerc.mongodb.net/?retryWrites=true&w=majority&appName=Ex8-
TieuPhuong';
mongoose
    .connect(mongoURI)
    .then(() => console.log('Connected to MongoDB'))
    .catch(err => console.error('Error connecting to MongoDB', err));

// Swagger UI
app.use('/api-docs', swaggerUi.serve, swaggerUi.setup(specs));

// Routes
app.use('/api/SensorDatas', sensorRoutes);
```

```
app.use('/api/Users', userRoutes);

// Server
const PORT = process.env.PORT || 3000;
app.listen(PORT, () => console.log(`Server running on
http://localhost:${PORT}`));
```

Running the Web API

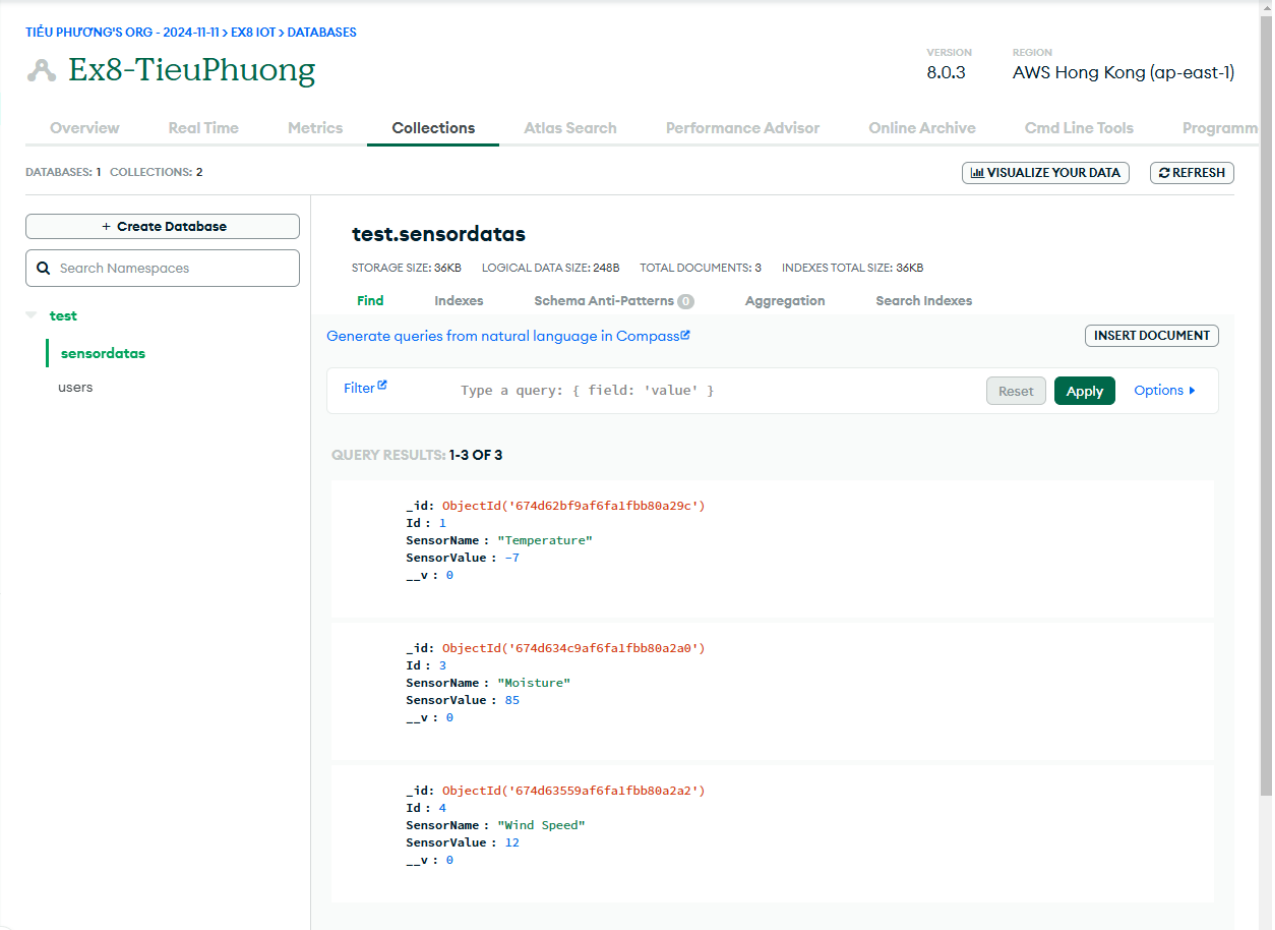


Server is running on port 3000.

Testing the API

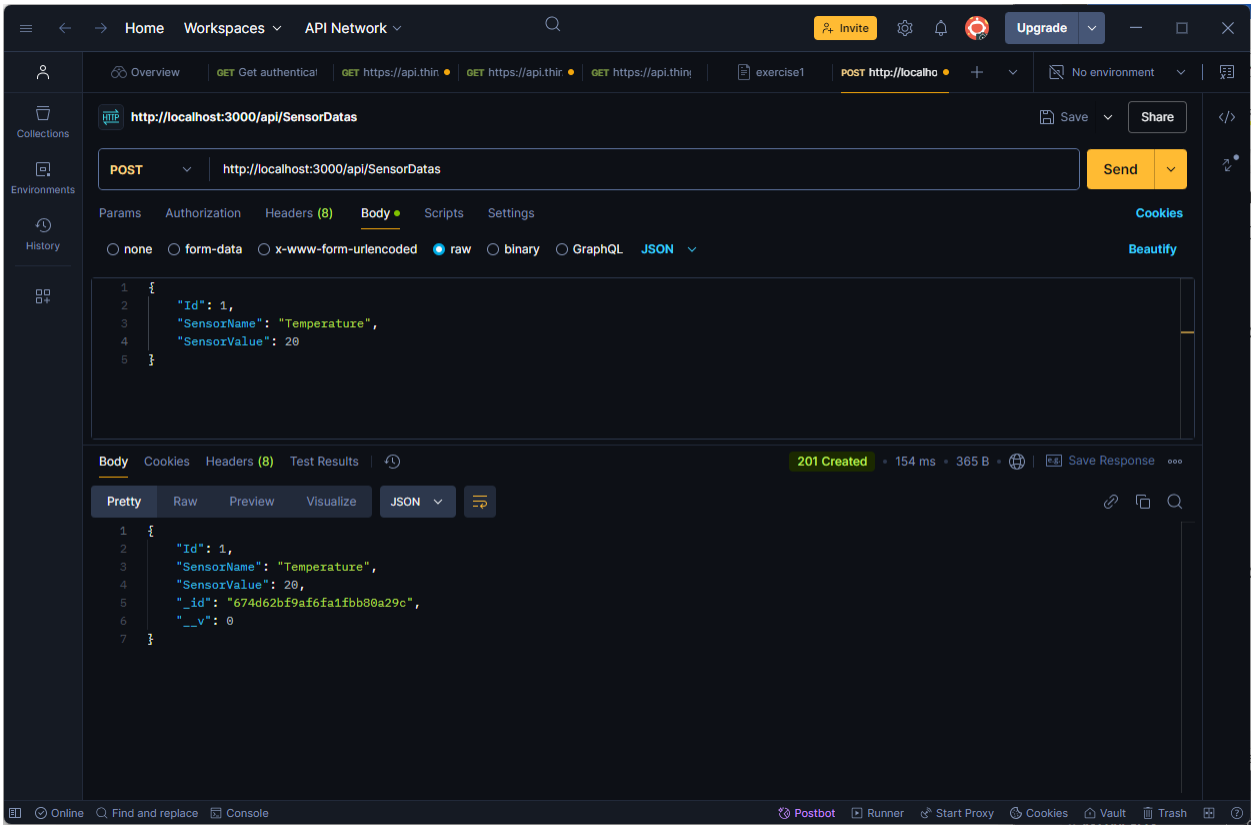
Note: MongoDB comes with a default `_id` field, auto generated string (not incremental). For this exercise, I create another, my own `Id` field that serves as the id for the users and sensor data.

SensorDatas

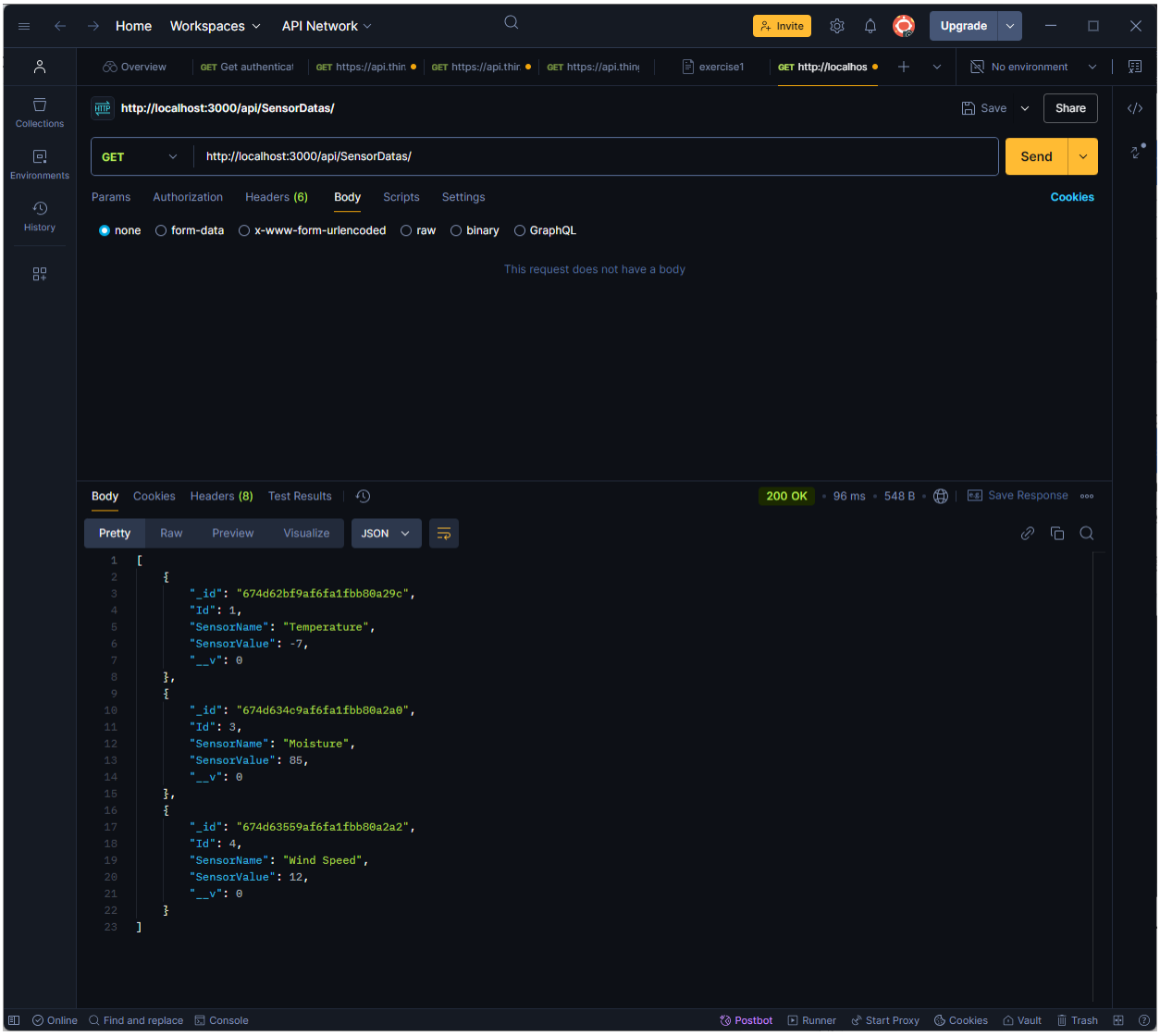


SensorDatas, as shown in MongoDB Atlas.

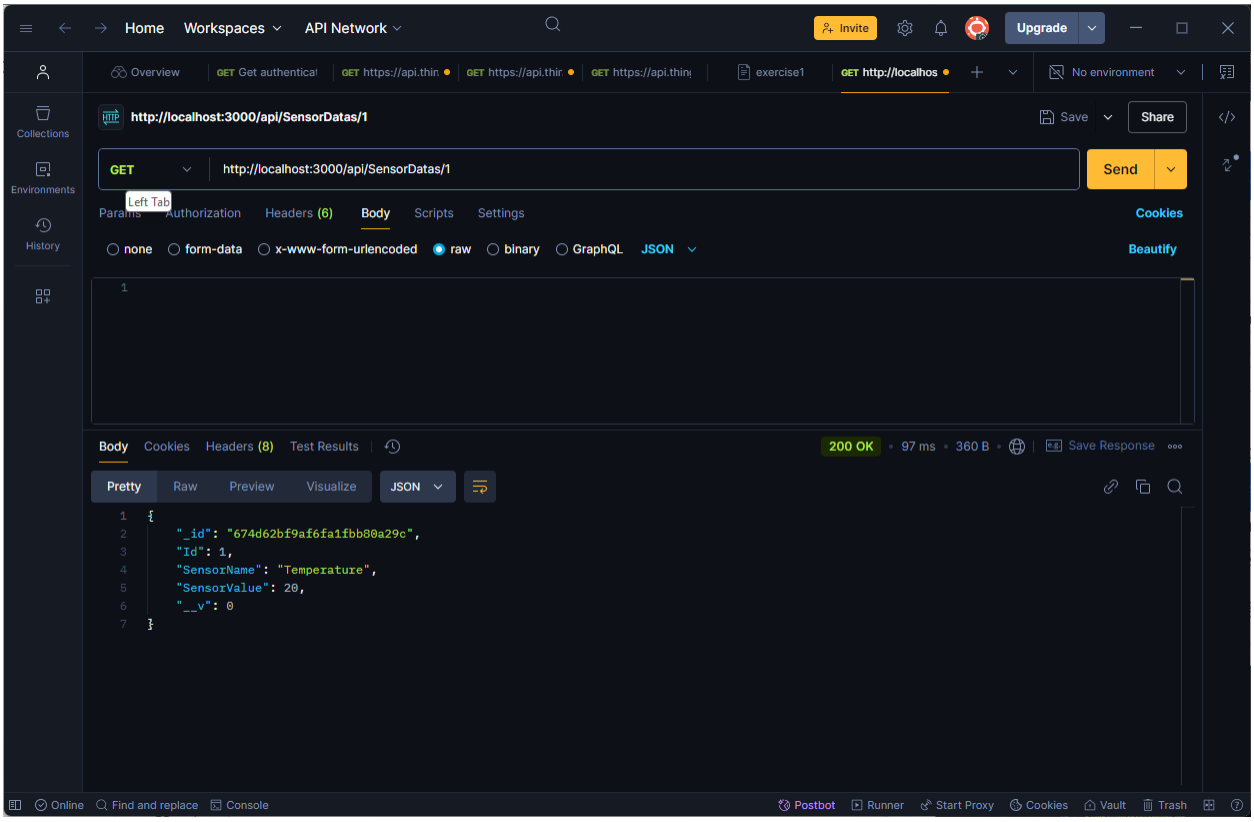
POST /api/SensorDatas



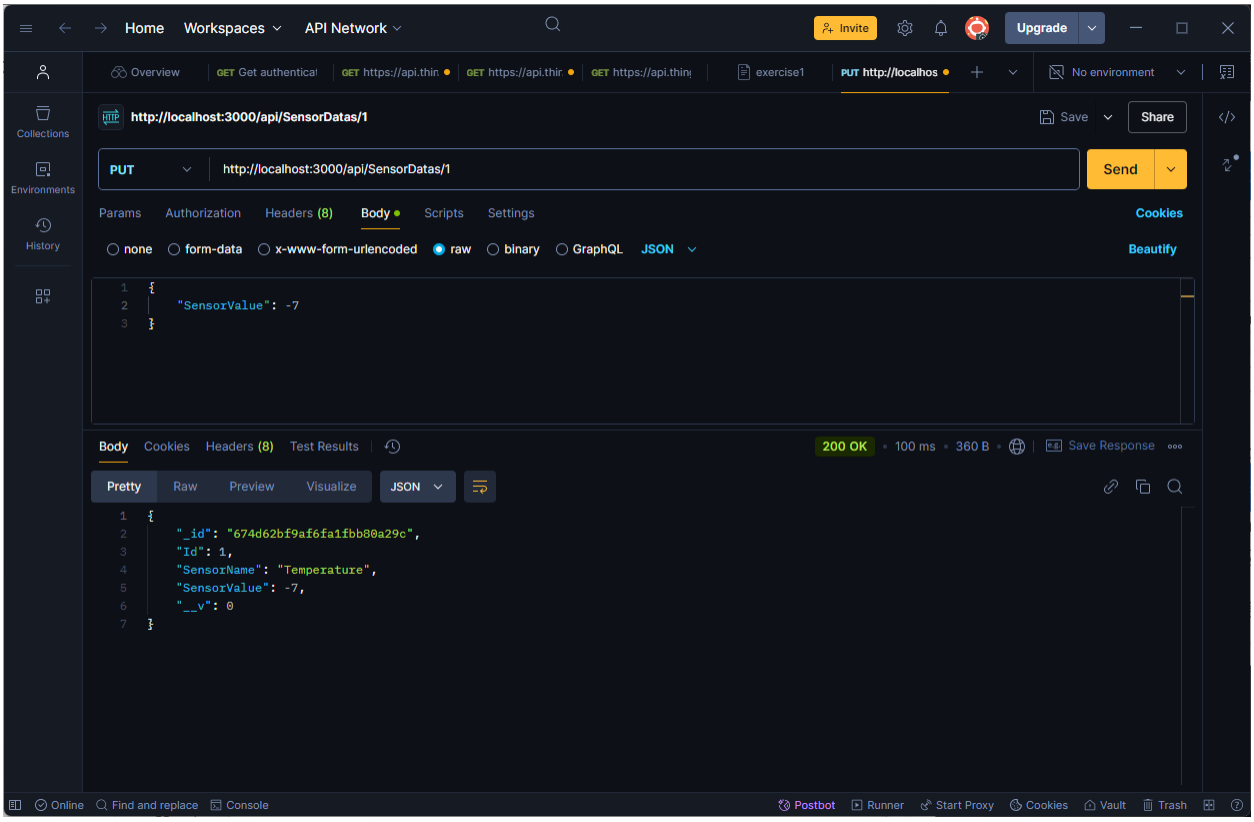
GET /api/SensorDatas



GET /api/SensorDatas/{id}

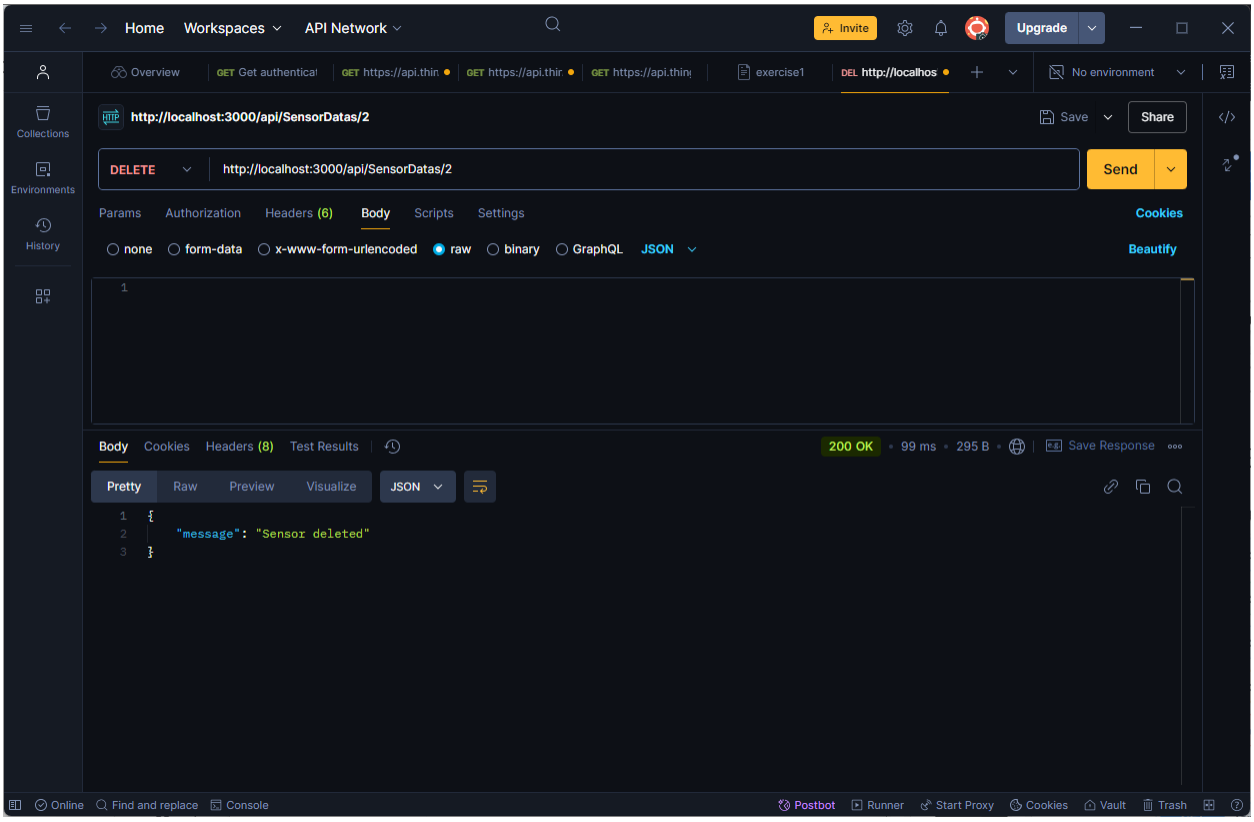


PUT /api/SensorDatas/{id}

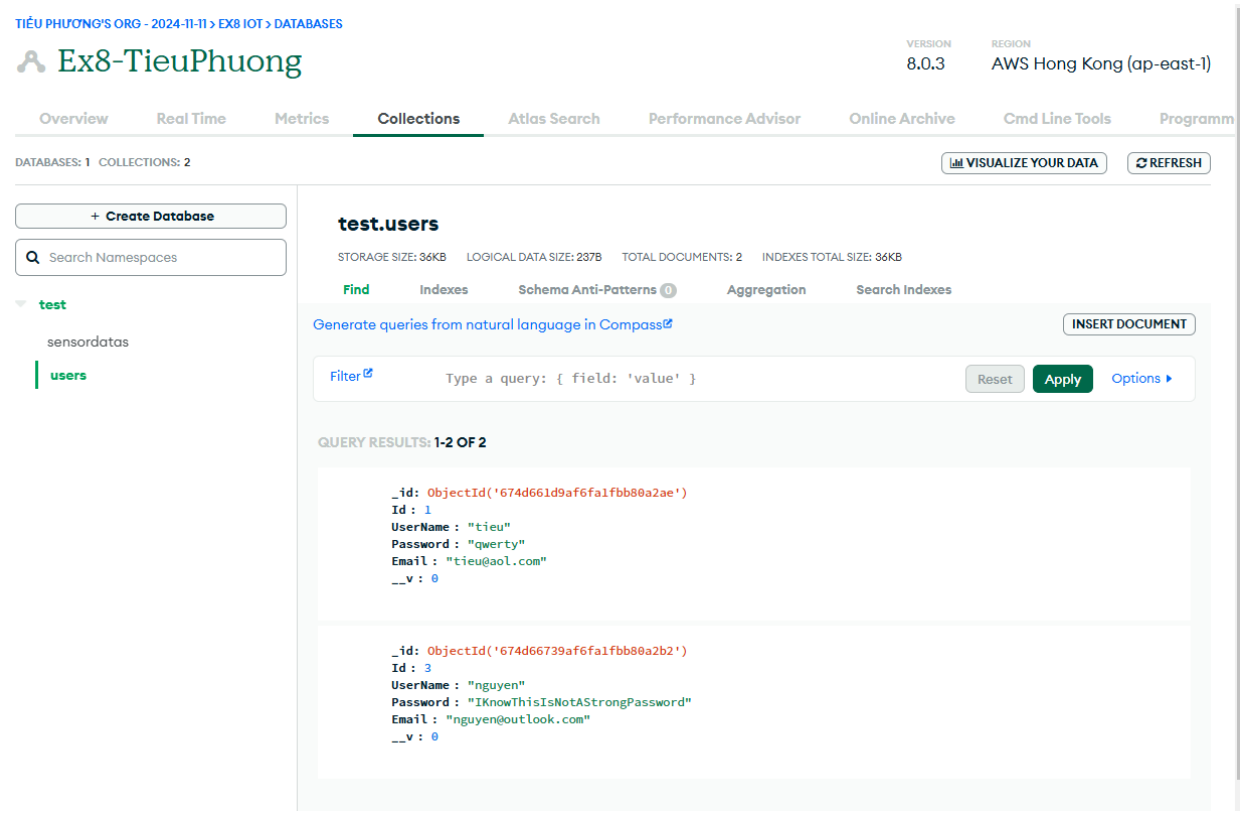


Change temperature value, for example.

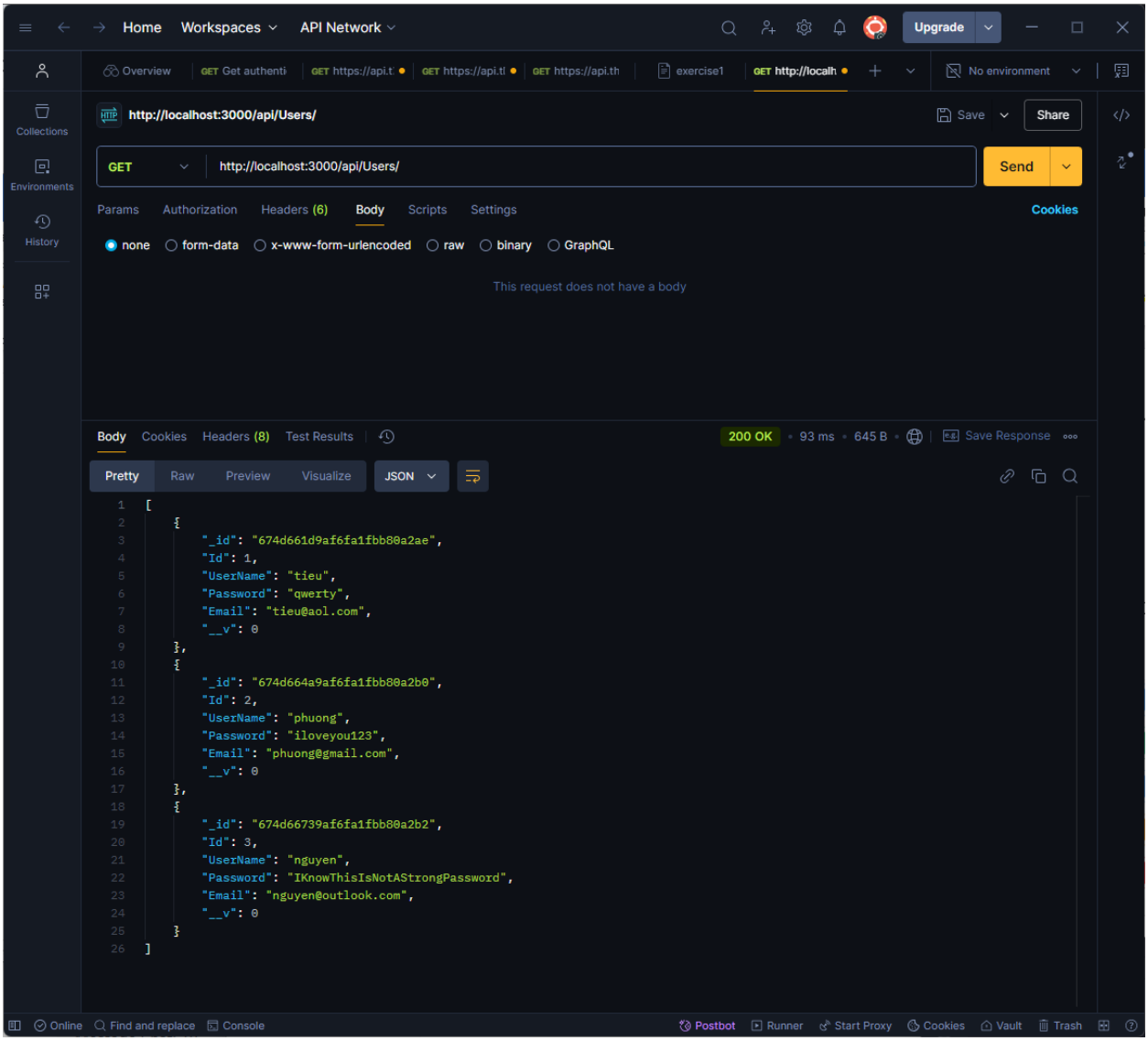
DELETE /api/SensorDatas/{id}



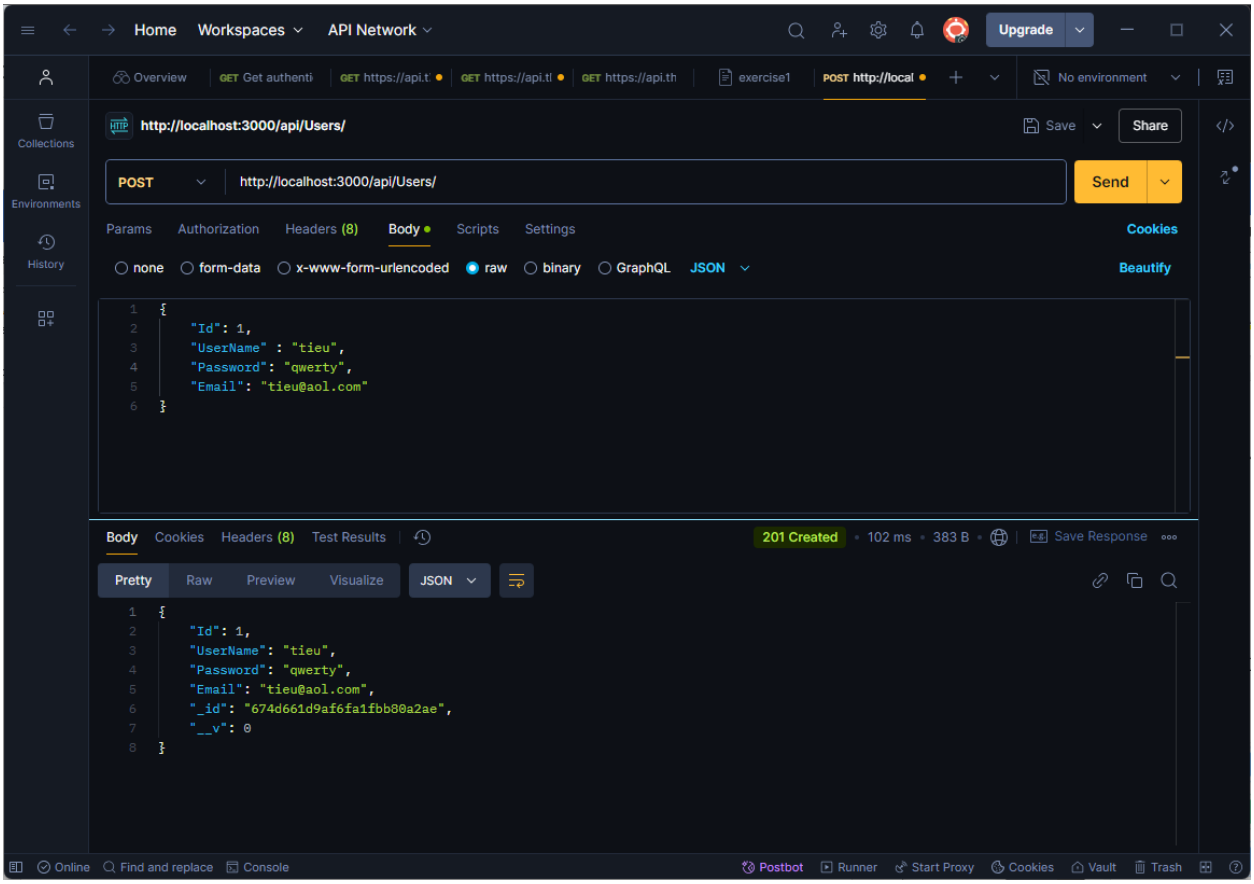
Users


TIÊU PHƯƠNG'S ORG - 2024-11-11 > EX8 IOT > DATABASES
Ex8-TieuPhuong
VERSION
8.0.3
REGION
AWS Hong Kong (ap-east-1)
Overview
Real Time
Metrics
Collections
Atlas Search
Performance Advisor
Online Archive
Cmd Line Tools
Programm
DATABASES: 1 COLLECTIONS: 2
VISUALIZE YOUR DATA
REFRESH
+ Create Database
Search Namespaces
test
sensordatas
users
test.users
STORAGE SIZE: 36KB
LOGICAL DATA SIZE: 237B
TOTAL DOCUMENTS: 2
INDEXES TOTAL SIZE: 36KB
Find
Indexes
Schema Anti-Patterns 0
Aggregation
Search Indexes
Generate queries from natural language in Compass
INSERT DOCUMENT
Filter
Type a query: { field: 'value' }
Reset
Apply
Options
QUERY RESULTS: 1-2 OF 2
_id: ObjectId('674d661d9af6fa1fbb80a2ae')
Id : 1
UserName : "tieu"
Password : "qwerty"
Email : "tieu@aol.com"
__v : 0
_id: ObjectId('674d66739af6fa1fbb80a2b2')
Id : 3
UserName : "nguyen"
Password : "IKnowThisIsNotAStrongPassword"
Email : "nguyen@outlook.com"
__v : 0

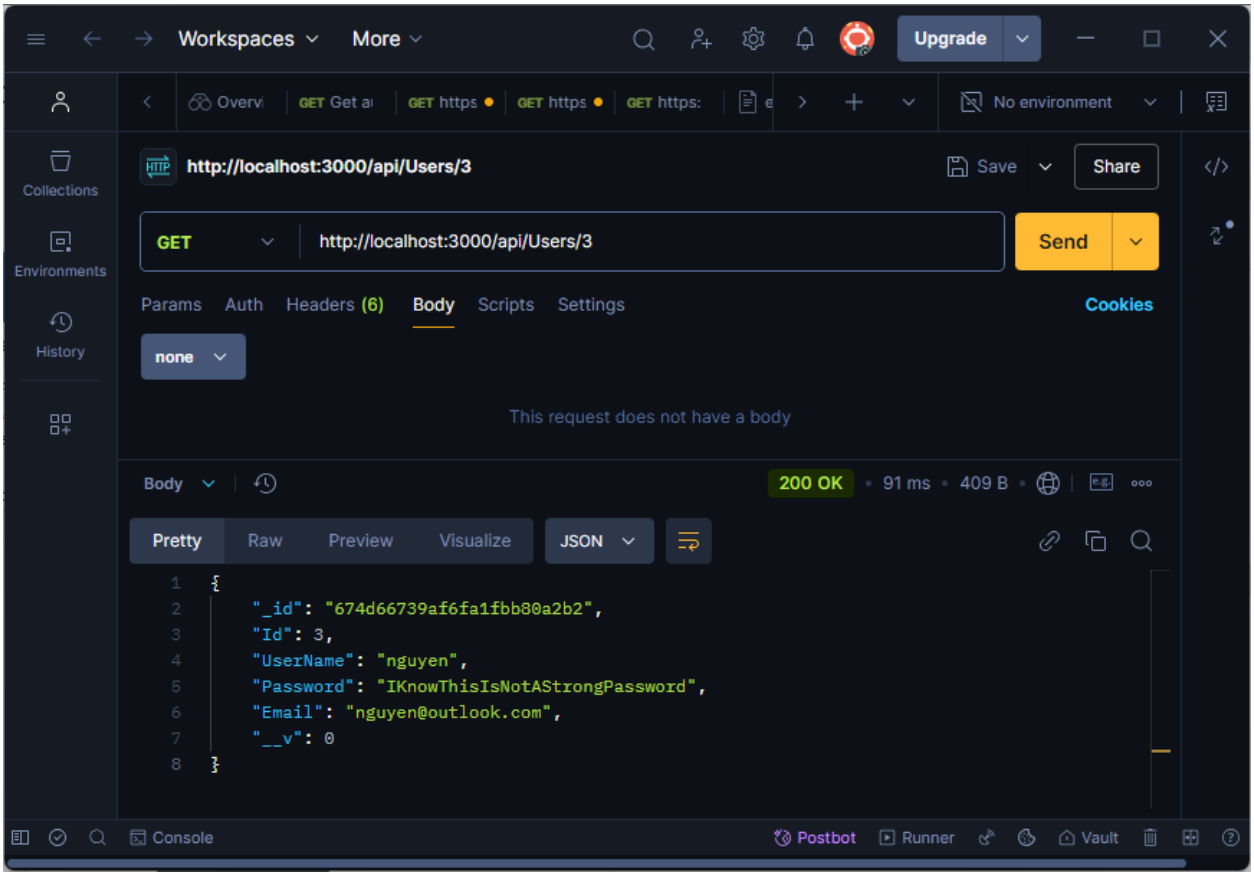
GET /api/Users



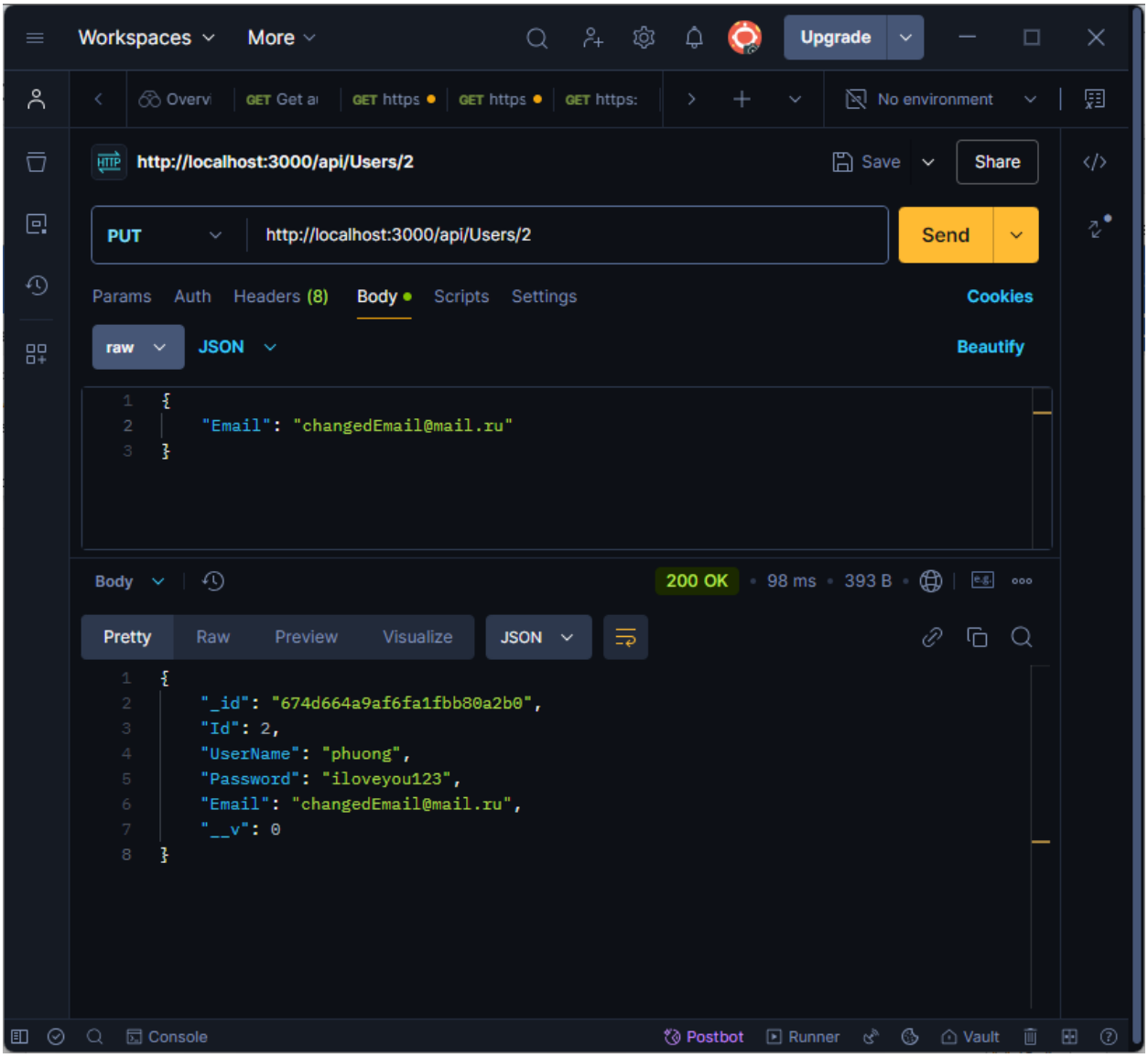
POST /api/Users



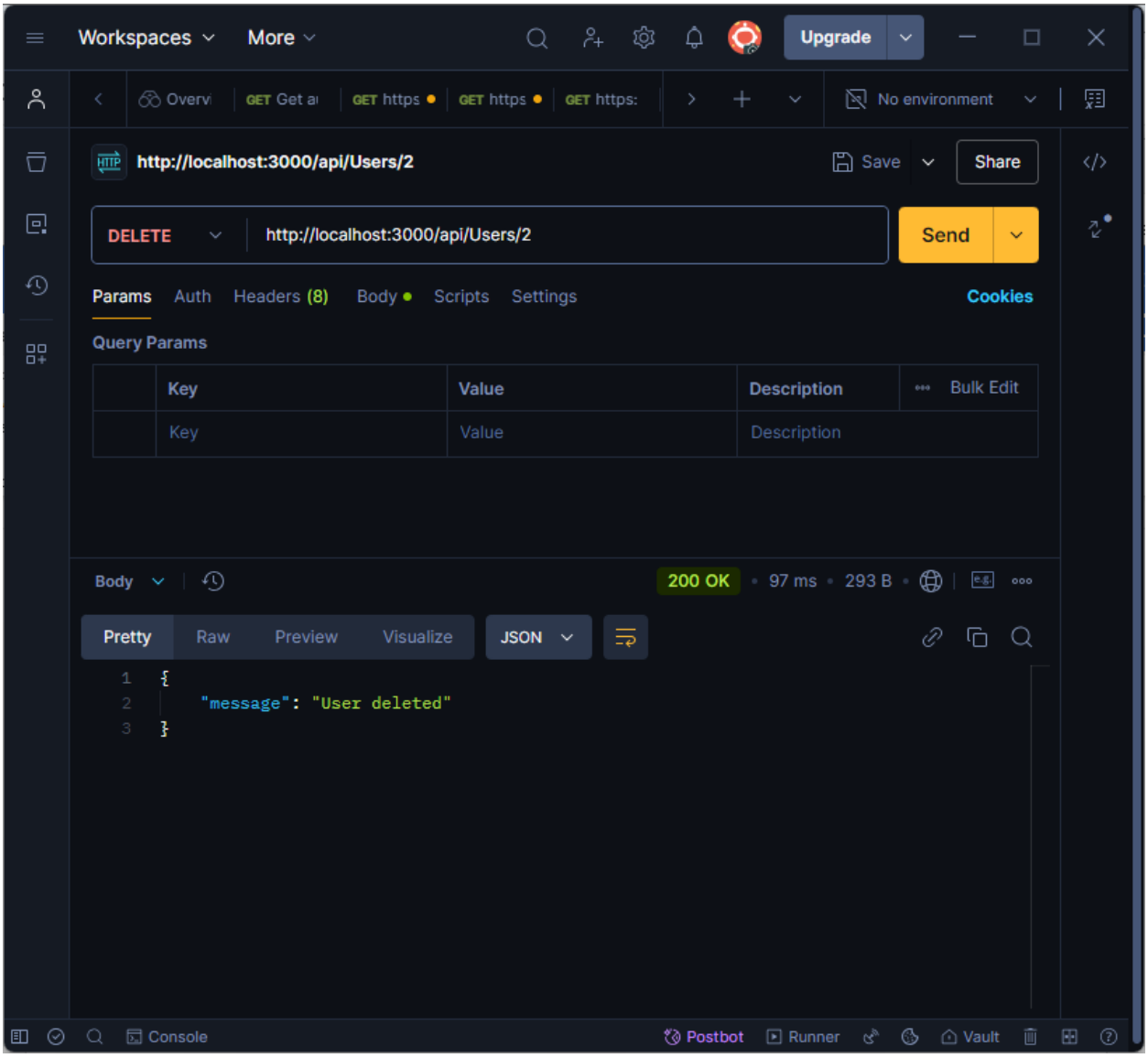
GET /api/Users/{id}



PUT /api/Users/{id}



DELETE /api/Users/{id}



THE END